

Case-II: Lifestyle and Sleeping Quality

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1 Case study sleep (task 2)

1.1 Part 1

```
old<-par(no.readonly = TRUE)

coljack<- matrix(c("#33ccff","#33ffbb","#33ff33","#99ff33","#ff9933","#ff3333","#ff3399","#f

#Reading the file
sleep<-read.csv("sleep.csv")
head(sleep)
```

	gender	age	occupation	sleep_duration	sleep_quality	stress_level
1	Male	27	Software Engineer	6.1	6	6
2	Male	28	Doctor	6.2	6	8

3	Male	28	Doctor	6.2	6	8
4	Male	28	Sales Representative	5.9	4	8
5	Male	28	Sales Representative	5.9	4	8
6	Male	28	Software Engineer	5.9	4	8
	daily_steps	sleep_disorder				
1	4200	None				
2	10000	None				
3	10000	None				
4	3000	Sleep Apnea				
5	3000	Sleep Apnea				
6	3000	Insomnia				

```
#types of data and sample size
str(sleep)
```

```
'data.frame': 374 obs. of 8 variables:
 $ gender      : chr  "Male" "Male" "Male" "Male" ...
 $ age         : int   27 28 28 28 28 28 29 29 29 ...
 $ occupation   : chr  "Software Engineer" "Doctor" "Doctor" "Sales Representative" ...
 $ sleep_duration: num  6.1 6.2 6.2 5.9 5.9 5.9 6.3 7.8 7.8 7.8 ...
 $ sleep_quality: int   6 6 6 4 4 4 6 7 7 7 ...
 $ stress_level : int   6 8 8 8 8 8 7 6 6 6 ...
 $ daily_steps  : int  4200 10000 10000 3000 3000 3000 3500 8000 8000 8000 ...
 $ sleep_disorder: chr  "None" "None" "None" "Sleep Apnea" ...
```

```
#changing tyoes of data to Factor for easier summary
sleep$occupation <-factor(sleep$occupation)
sleep$sleep_disorder <- factor(sleep$sleep_disorder)
```

```
#summary for main data
summary(sleep)
```

gender	age	occupation	sleep_duration
Length:374	Min. :27.00	Nurse :73	Min. :5.800
Class :character	1st Qu.:35.25	Doctor :71	1st Qu.:6.400
Mode :character	Median :43.00	Engineer :63	Median :7.200
	Mean :42.18	Lawyer :47	Mean :7.132
	3rd Qu.:50.00	Teacher :40	3rd Qu.:7.800
	Max. :59.00	Accountant:37	Max. :8.500
		(Other) :43	

sleep_quality	stress_level	daily_steps	sleep_disorder
Min. :4.000	Min. :3.000	Min. : 3000	Insomnia : 77
1st Qu.:6.000	1st Qu.:4.000	1st Qu.: 5600	None :219
Median :7.000	Median :5.000	Median : 7000	Sleep Apnea: 78
Mean :7.313	Mean :5.385	Mean : 6817	
3rd Qu.:8.000	3rd Qu.:7.000	3rd Qu.: 8000	
Max. :9.000	Max. :8.000	Max. :10000	

```
#distribution of age, sleep duration, stress level, daily steps
```

```
sapply(sleep[,c("age","sleep_duration","stress_level","daily_steps")], FUN=sd)
```

age	sleep_duration	stress_level	daily_steps
8.6731335	0.7956567	1.7745264	1617.9156791

```
#Types of Jobs at a study and length
```

```
unique(sleep$occupation)
```

```
[1] Software Engineer Doctor Sales Representative
[4] Teacher Nurse Engineer
[7] Accountant Scientist Lawyer
[10] Salesperson Manager
11 Levels: Accountant Doctor Engineer Lawyer Manager ... Teacher
```

```
length(unique(sleep$occupation))
```

```
[1] 11
```

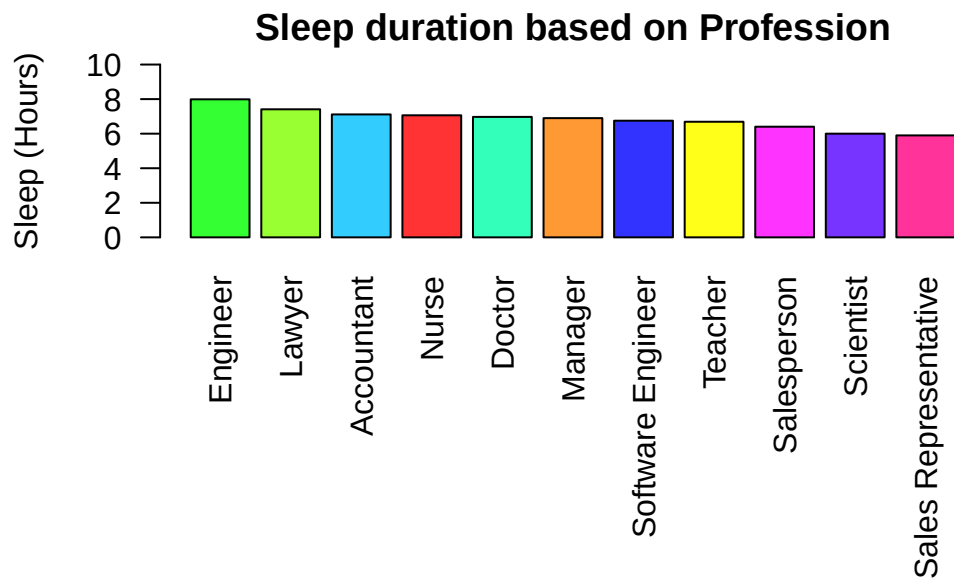
```
sleepocdu<-aggregate( sleep_duration ~ occupation, data = sleep, FUN = mean)
sleepocdu<-cbind(sleepocdu,coljack)
sleepocdu <- sleepocdu[order(-sleepocdu$sleep_duration),]
print(sleepocdu)
```

	occupation	sleep_duration	coljack
3	Engineer	7.987302	#33ff33
4	Lawyer	7.410638	#99ff33
1	Accountant	7.113514	#33ccff
6	Nurse	7.063014	#ff3333

2	Doctor	6.970423	#33ffbb
5	Manager	6.900000	#ff9933
10	Software Engineer	6.750000	#3333ff
11	Teacher	6.690000	#ffff1a
8	Salesperson	6.403125	#ff33ff
9	Scientist	6.000000	#7733ff
7	Sales Representative	5.900000	#ff3399

```
par(mai = c(0.1, 0.4, 0.1, 0.2), mar = c(11, 4, 2, 2))
```

```
bpsd <- barplot(
  sleepocdu$sleep_duration, col = sleepocdu$coljack, main = "Sleep duration based on Profes
)
```



1.2 Part 2

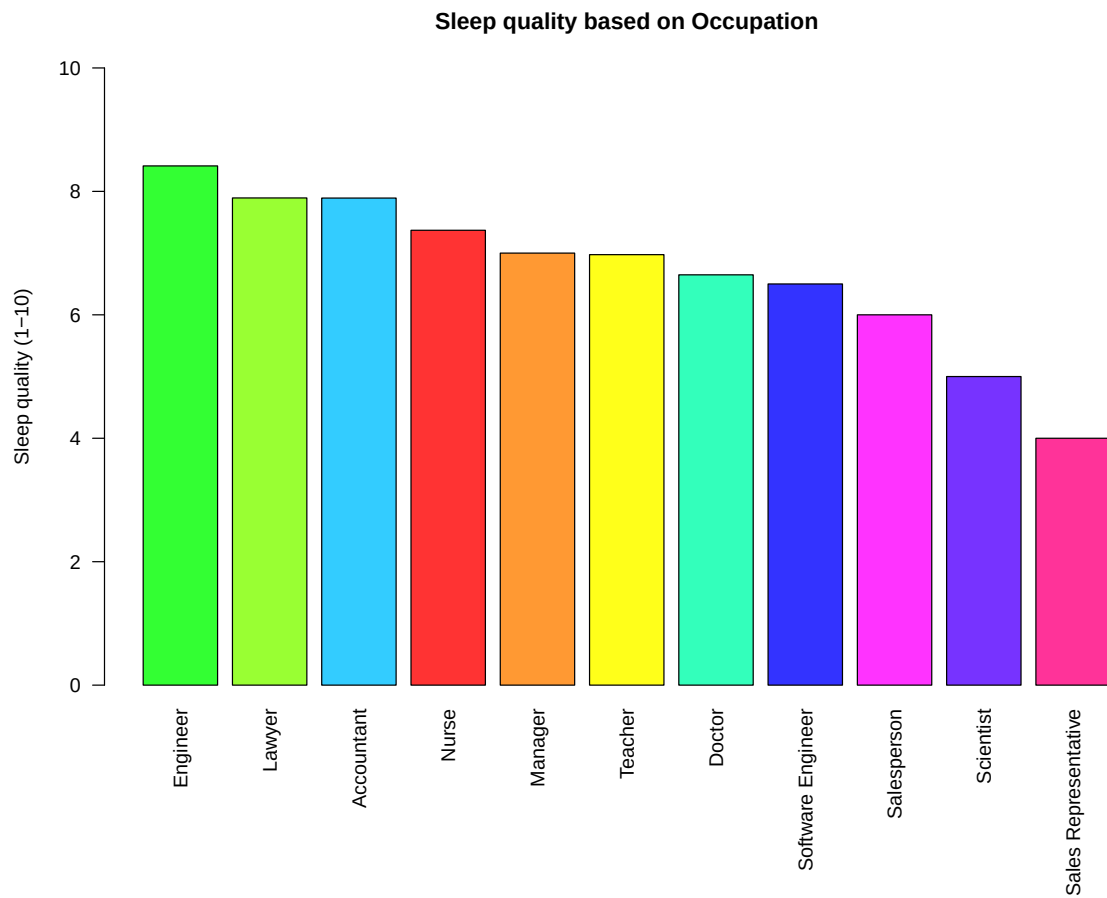
1.2.1 2a Relation between Occupation and Stress level, Sleep quality

1.2.1.1 Sleep Quality and Occupation

```
sleepocqw<-aggregate( sleep_quality ~ occupation, data = sleep, FUN = mean)
sleepocqw<-cbind(sleepocqw,coljack)
sleepocqw <- sleepocqw[order(-sleepocqw$sleep_quality),]
print(sleepocqw[,c("occupation","sleep_quality")])
```

	occupation	sleep_quality
3	Engineer	8.412698
4	Lawyer	7.893617
1	Accountant	7.891892
6	Nurse	7.369863
5	Manager	7.000000
11	Teacher	6.975000
2	Doctor	6.647887
10	Software Engineer	6.500000
8	Salesperson	6.000000
9	Scientist	5.000000
7	Sales Representative	4.000000

```
par(mar = c(10, 4, 4, 2))
barplot(sleepocqw$sleep_quality,
        names.arg = sleepocqw$occupation,
        col = sleepocqw$coljack,
        main = "Sleep quality based on Occupation",
        axis.lty = 0,
        las=2,
        ylim = c(0,10),
        ylab = "Sleep quality (1-10)"
)
```



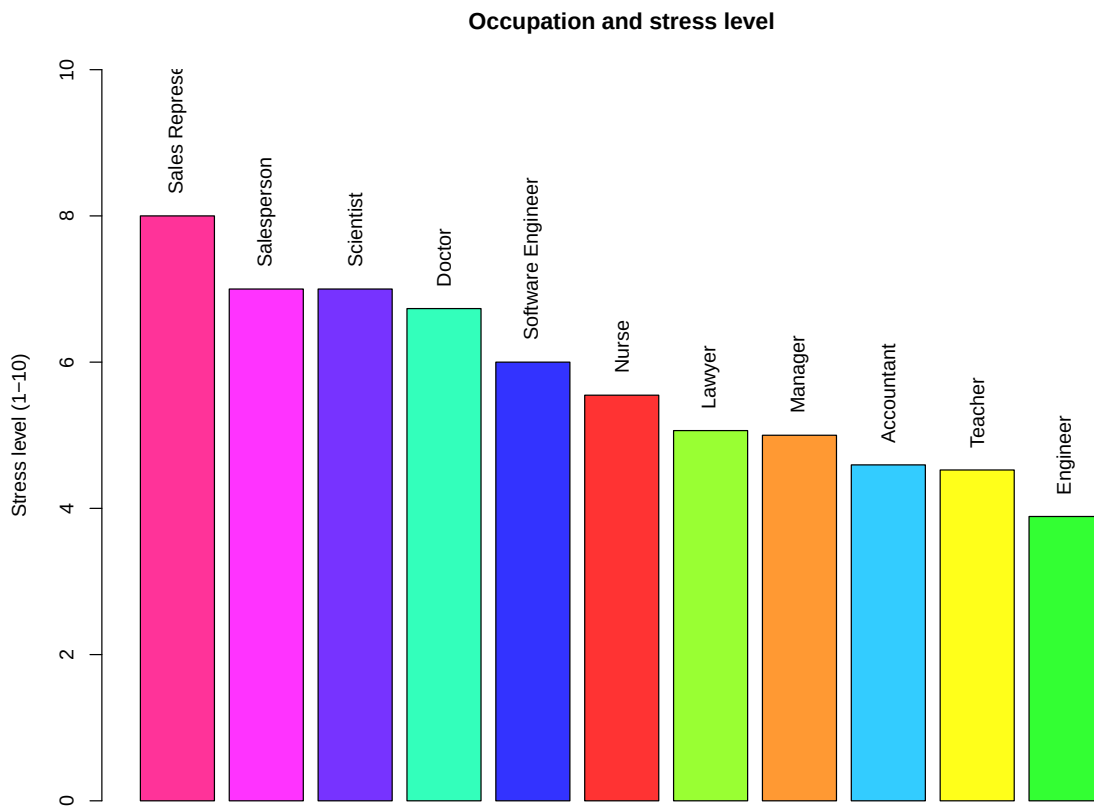
1.2.1.2 Stress Level and Occupation

```
sleepocst<-aggregate(stress_level ~ occupation, data = sleep, FUN = mean )
sleepocst<-cbind(sleepocst,coljack)
sleepocst<-sleepocst[order(-sleepocst$stress_level),]
print(sleepocst[,c("occupation","stress_level")])
```

	occupation	stress_level
7	Sales Representative	8.000000
8	Salesperson	7.000000
9	Scientist	7.000000
2	Doctor	6.732394
10	Software Engineer	6.000000
6	Nurse	5.547945

4	Lawyer	5.063830
5	Manager	5.000000
1	Accountant	4.594595
11	Teacher	4.525000
3	Engineer	3.888889

```
bpst<-barplot(sleepocst$stress_level, col = sleepocst$coljack, main = "Occupation and stress
text(x = bpst, y = sleepocst$stress_level + 0.3, labels = sleepocst$occupation, srt = 90,adj
)
```



1.2.2 2b Sleep disorder

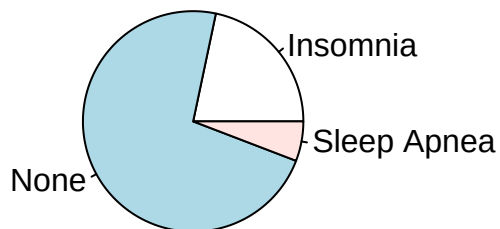
```
table(sleep$sleep_disorder)
```

Insomnia	None	Sleep Apnea
77	219	78

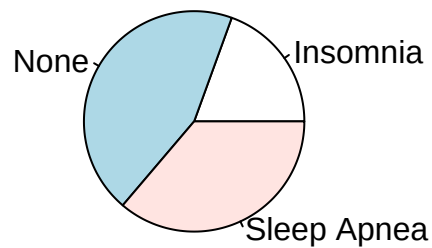
```
gender_w_dis<-table(sleep$gender, sleep$sleep_disorder)
```

```
par(mfrow=c(1,2),mar= c(2,2,2,4))  
pie(gender_w_dis["Male", ],main = "Male")  
  
pie(gender_w_dis["Female", ],main = "Female")
```

Male



Female



```
par(old)
```

Sleep Apnea is most common overall (excluding no disorder)

Most common in Female is Apnea Most common in Male is Insomnia

1.2.3 2c Combination of Occupation gender Quality

```
occ_gen_qual <- aggregate(sleep_quality ~ occupation + gender, data = sleep, FUN = mean)
counts <- aggregate(sleep_quality ~ occupation + gender, data = sleep, FUN = length)
names(counts)[3] <- "n"
occ_gen_qual <- merge(occ_gen_qual, counts, by = c("occupation", "gender"))
occ_gen_qual <- occ_gen_qual[order(-occ_gen_qual$sleep_quality), ]
occ_gen_qual
```

	occupation	gender	sleep_quality	n
3	Doctor	Female	9.000000	2
5	Engineer	Female	9.000000	32
2	Accountant	Male	8.000000	1
8	Lawyer	Male	7.933333	45
1	Accountant	Female	7.888889	36
6	Engineer	Male	7.806452	31
10	Nurse	Female	7.369863	73
15	Teacher	Female	7.114286	35
7	Lawyer	Female	7.000000	2
9	Manager	Female	7.000000	1
4	Doctor	Male	6.579710	69
14	Software Engineer	Male	6.500000	4
12	Salesperson	Male	6.000000	32
16	Teacher	Male	6.000000	5
13	Scientist	Female	5.000000	4
11	Sales Representative	Male	4.000000	2

```
sleep_male<-sleep[sleep$gender %in% c("Male"),]
sleep_female<-sleep[sleep$gender %in% c("Female"),]
```

Lowest Male sleep quality is for Sales Representatives

Lowest Female sleep quality is for Scientist

```
sleepocst_Male<-aggregate(stress_level ~ occupation, data = sleep_male, FUN = mean )
sleepocst_Male<-sleepocst_Male[order(-sleepocst_Male$stress_level),]
print(sleepocst_Male)
```

	occupation	stress_level
5	Sales Representative	8.000000

6	Salesperson	7.000000
2	Doctor	6.840580
8	Teacher	6.200000
1	Accountant	6.000000
7	Software Engineer	6.000000
4	Lawyer	5.044444
3	Engineer	4.806452

Male Sales Representatives has highest stress

```
sleepocst_Female<-aggregate(stress_level ~ occupation, data = sleep_female, FUN = mean )
sleepocst_Female<-sleepocst_Female[order(-sleepocst_Female$stress_level),]
print(sleepocst_Female)
```

	occupation	stress_level
7	Scientist	7.000000
6	Nurse	5.547945
4	Lawyer	5.500000
5	Manager	5.000000
1	Accountant	4.555556
8	Teacher	4.285714
2	Doctor	3.000000
3	Engineer	3.000000

Female Scientist has highest stress

```
print(cor(sleep$sleep_quality,sleep$stress_level))
```

```
[1] -0.898752
```

There is a negative correlation between stress level and sleep quality meaning the more stress a person have the worse is there sleep quality. Also basing that there is a similar correlation in separate female and male group i think that there is enough evidence to say there can be a strong correlation.

1.3 Part 3

```
sleep_male_insomnia<-sleep_male[sleep_male$sleep_disorder %in% c("Insomnia"),]
print(length(sleep_male_insomnia$gender)/length(sleep_male$gender)*100)
```

```
[1] 21.69312
```

Wrong 21.69 percent of males struggle from Insomnoa

```
print(mean(sleep_male_insomnia$daily_steps))
```

```
[1] 5812.195
```

Wring the average daily steps by males with insomnia is 5812

1.4 Part 4

1.4.1 a)

1.4.1.1 1.

```
sleep_insomnia<-sleep[sleep$sleep_disorder %in% c("Insomnia"),]
summary(sleep_insomnia$age)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
28.00	43.00	44.00	43.52	45.00	53.00

```
par(mfrow=c(1,2),mar= c(5,1,5,1))
```

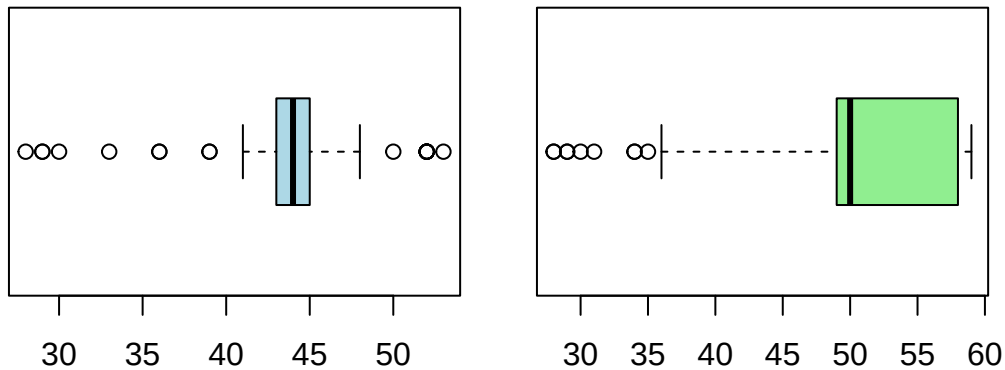
```
boxplot(sleep_insomnia$age, horizontal = TRUE, main = "Age of people with Insomnia", col= "1")
```

```
sleep_apnea<-sleep[sleep$sleep_disorder %in% c("Sleep Apnea"),]
print(summary(sleep_apnea$age))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
28.00	49.00	50.00	49.71	57.75	59.00

```
boxplot(sleep_apnea$age, horizontal = TRUE, main = "Age of people with Sleep Apnea", col = "lightblue")
```

Age of people with Insomnia vs Age of people with Sleep Apnea



1.4.2 a)

1.4.2.1 2.

```
sleep_insomnia<-sleep[sleep$sleep_disorder %in% c("Insomnia"),]
summary(sleep_insomnia$sleep_duration)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
5.90	6.40	6.50	6.59	6.60	8.30

```
par(mfrow=c(1,2),mar= c(5,1,5,1))
```

```
boxplot(sleep_insomnia$sleep_duration, horizontal = TRUE, main = "Duration of sleep with Insomnia")
```

```
sleep_apnea<-sleep[sleep$sleep_disorder %in% c("Sleep Apnea"),]
print(summary(sleep_apnea$sleep_duration))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
5.800	6.100	6.800	7.032	8.100	8.200

```
boxplot(sleep_apnea$sleep_duration, horizontal = TRUE, main = "Duration of sleep with Sleep Ap
```

Duration of sleep with Insomniration of sleep with Sleep Apr

